Test

MCQ: **20x1=20**

1. Showing only important information and hiding background details is

a) Encapsulation b) Data Abstraction

c) Inheritance d) Polymorphism

2. WORA is feature of JAVA. It stands for

a) Write once run anywhere b) Write on runtime assistance

c) Write Object Run Anywhere d) None of the above

3. Amount of memory occupied by 15.6 and '\t'

a) 8 bytes and 2 bytes b) 8 bits and 2 bits

c) 16 bytes and 4 bytes d) 8 bytes and 4 bytes

4)
```
for(boolean i=true; i ; i= !i){
    System.out.println( !i ? "HI": "Hello");
}
```

a) Hi b) Hello

c) Infinite loop d) Error

5) Identify the operator that gets the highest precedence or is executed first while evaluating the expression:

5-5+ * 4 %3/2

a) - b) *

c) + d) %

6) Which of the following is a valid keyword

a) switch b) system

c) string d) Boolean

7. System.out.print(Math.floor(-3-2) + Math.round(4.6));

a) 10.0 b) 10

c) 0 d) 0.0

Which of the following correctly converts String to int

String s=”121”;

a) (int) s

b) String.toInteger(s)

c) Integer.pareInt(s)

d) Integer.parseint(s)

9. int a= new Integer(10); is an example of

a) Coercion

b) Unboxing

c) Boxing

d) None

10. Integer ob= 20;

a) Autoboxing

b) Unboxing

c) Boxing

d) None

11. which is not true

a) switch case can be nested

b) switch case can use String as case labels

c) switch case can use float as labels

d) default is optional in switch case

12. String s= “APPLE”; System.out.printlnl(s.charAt(4) *2);

a) 138

b) Error

c) 136

d) 140

13) which of the following loop won’t run even once

a) for(int i=0; i<=0 i++)

b) for(int i=0;i<=0; i++);

b) for(int i=1; i<1; i++)

d) for(int i=1; i>=1; i++)

14) int a[]={2,4,5,6,7,8}; int a= a[2+2] * a[2]+a[2]; a=?

a) 40

b) 49

c) 125

d) 5

****** Default values are only assigned to class members and arrays.***

15) Method prototype of a function that display that takes boolean and returns a character

}

a) boolean display(char ch)

b) char display(boolean b)

c) char display(false)

d) None

16. Which of the following skips some codes and goes to the next iteration

a) break
b) continue

c) return
d) System.exit(0);

17. default value of char is

a) ‘ ‘
b) 32

c) ‘\u0000’
d) “\U000’

18) Which of the following checks if a character is lowercase or not

a) Character.islowercase(char)
b) char.isLowerCase()

c) char.isLowerCase(char)
d) Character.isLowerCase(char)

19. Assertion: a[a.length-1] is valid

Reason: An array of N size has N -1 elements

Assertion and reason are

a) true and false
b) false and true

c) false and false
d) true and true

20. Which one is correct way to swap two variables without using a third variable

a) a=a+b; a=a-b; b=b-a;
b) a=a+b; a=a-b; b=a-b;

b) a=a+b; b=a-b; a=a-b;
d) a=b, b=a;

SAQ. Answer the following (Showing dry run is compulsory for snippets) **10x2=20**

1. Convert to switch case

```
if( a==1 || a==2) {
System.out.println("Hello");
}else{
System.out.println("Hello2");
}
```

2. find the value of x and y; x=2, y=3;

x*= x++ + ++x – y-- - --y + y%x;

3. Write the following expression in JAVA.

k= $\sqrt[4]{|x-y|}$

4.
```
public class SwitchCaseExample {
    public static void main(String[] args) {
        int num = 2;
        switch (num) {
            case 1:
                System.out.println("One");
            case 2:
                System.out.println("Two");
            case 3:
                System.out.println("Three");
            case 4:
                System.out.println("Four");
                break;
            default:
                System.out.println("Default");
        }
    }
}
```

find output.

5. Find output and the number of times the loop executes

```
int n=1234;
     while(true){
             int d=n%10;
             System.out.println(d);
             if(d*3==6)break;
             n/=10;
     }
```

6.
```
String s1= "ASCII", s2= "ascii";
System.out.println(s1.compareTo(s2));
System.out.println(s1.compareToIgnoreCase(s2));
```

7. Predict the output

```
char ch1= 'A', ch2= 'B';
System.out.println(ch1+ch2);
System.out.println((char)(ch1+ch2-66);
```

8. class Error{

```
class Error{
    public static void main(String[] args) {
        int result = Math.pow(2, 3.5);
        int roundedResult = Math.round(result);
        System.out.println("Result: " + result);
        System.out.println("Rounded Result: " + roundedResult);
    }
}
```

find the error. what type of error is it ? correct it.

9.
int a[][]={ {1,2,3}, {3,4,5}, {4,5,6}};
find the order of the array. also find a[1][0]+a[0][1];

10. Convert to ternary:

```
if(a==1) a=1;
else if (a==2) a=2;
else a==3;
```

Section B

3. Define a class Bill that calculates the telephone bill of a consumer with the following description:

Data Members	**Purpose**
int bno	bill number
String name	name of consumer
int call	no. of calls consumed in a month
double amt	bill amount to be paid by the person

Member Methods	**Purpose**
Bill()	constructor to initialize data members with default initial value
Bill(...)	parameterised constructor to accept billno, name and no. of calls consumed
Calculate()	to calculate the monthly telephone bill for a consumer as per the table given below
Display()	to display the details

Units consumed	**Rate**
First 100 calls	₹0.60 / call
Next 100 calls	₹0.80 / call
Next 100 calls	₹1.20 / call
Above 300 calls	₹1.50 / call

Fixed monthly rental applicable to all consumers: ₹200

Create an object in the main() method and invoke the above functions to perform the desired task.

4. Write a JAVA Program to take an element from user and print its index using Binary Search

100, 95, 80, 70, 50, 40, 35, 20, 15, 5

5. Write a program to overload a function Overload() as follows

Overload(int n) to print

if n==3 then

1 2 3

1 2 3

1 2 3

Overload (int n, int x)

find S= x/ (2k) - x/(3k) + x/(4k) - x/(5k) ..... upto n

here x is given by user and k varies from 1 to n

6. Write a program to take a String and replace each vowel with next consonant and remove each consonant

7. Take a matrix of order 4x4 from user find sum of each row, each column, boundary elements, left diagonal and right diagonal

8. Take a number and check if its product of digits is equal to the factorial of the number or not